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ABSTRACTS

Circular Economy in the Electronics Sector: A Comparative Analysis and Framework of WEEE Policy and Practice in the EU and India

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The volume of e-waste is rising exponentially around the globe. Investigating how different nations manage Waste from Electrical and Electronic Components (WEEE) is crucial for designing policies that keep pace with technological innovation while preventing environmental degradation. In this study, we develop a WEEE Circularity and Governance Scoring Framework, drawing from the UN, the Basel Convention, and recent literature. It examines four key areas – legislative action, executive implementation, judicial mechanisms, and international cooperation. For cross-regional comparison, we selected two EU leaders in WEEE management (Netherlands, Austria), two relatively lower-performing EU states (Italy, Romania), and one major non-EU developing country (India), enabling performance evaluation across governance and development contexts. EU nations score higher than India, reflecting gaps in repairability mandates, informal sector integration, and consumer incentives. The approach provides a replicable tool for policymakers to identify targeted interventions and close governance score gaps.

System Dynamics Modelling of Data Center Emissions: Evidence from Nordic and North American Markets

Sean Hansed, Jack o'Grady, Pragyan Nandan Pandey, Noah Williams

This study bridges the gap between theoretical research on data centres and the applied development of sustainability policies, addressing the sector's rapid global expansion and shift into new markets. A systems model of data centre policy development is proposed and evaluated through a comparative analysis of the North American and Nordic markets. Primary data are used to examine how operators mitigate environmental externalities and meet sustainability commitments. Methodologically, behavioural patterns are analysed using Principal Component Analysis (PCA) and Data Envelopment Analysis (DEA) to assess policy effectiveness and efficiency trade-offs. Results demonstrate a distinct divergence between the two regions: North American data centres exhibit higher efficiency in data delivery, while Nordic facilities achieve greater efficiency in emissions optimisation. These findings highlight the inherent trade-off between operational and environmental objectives. The refined systems model provides a structured framework for understanding policy dynamics and offers a foundation for future empirical work with larger datasets.

A Qualitative Framework for Agro-PV Adoption in India: A System Dynamics Study

Chunendra Kumar, Bhakti Prasad, Aarchi P Kothari, Mohammad Usaid, Ankush Halba

This study investigates the multidimensional adoption of Agro-Photovoltaic (Agro-PV) systems in India by developing a qualitative framework using system dynamics methodology. Agro-PV, where PV modules are combined with agricultural activities, offers a two-fold benefit in terms of clean energy and sustainable agricultural solutions. Yet, its uptake has been influenced by the intricate relationship of technical, economic, and social factors in the Indian situation. The method adopted to solve this study is embedded in the Structured Modeling (SM) methodology and starts with the generation of separate Causal Loop Diagrams (CLDs) for each knowledge domain (technical, economic and social dimension: system design, crop yield and land optimization, investment costs, revenue streams, subsidies and employment, rural livelihoods, collective action, awareness and training). These models are relations among important variables, feedback processes, and possible 'balancing' or 'reinforcing' dynamics, have been described elsewhere. The last stage connects these initial models into a unified CLD, representing the dynamics of the drivers and inhibitors for Agro-PV. Here, we develop a conceptual framework to facilitate policy intervention, identify the leverage points for upscaling Agro-PV, and inform the design of implementation strategies addressing technical reliability, financial viability and social acceptability. In the end, this work will add to literature on sustainable technology diffuse in agriculture, with the set of learnings for faster promotion of Agro-PV in India.

The Impact of Variable Exclusion: How Missing Data Contributes to Bias in Alzheimer's Disease AI Models

Earlel Thiyagaratnam, Hawwa Khan

Alzheimer's disease (AD) is the most common cause of dementia, yet many cases go undiagnosed due to limited access to expensive brain scans and lab tests. This study investigated six datasets, demographic, medication, laboratory, examination, diet, and questionnaire data that could help identify AD. Using data from 1,785 participants in the US-representative National Health and Nutrition Examination Survey (NHANES) 2013–2014, we identified 105 individuals (5.9%) with memory test scores suggesting possible AD. We evaluated seven machine learning models using the medication features. For our first analysis, models that incorporated contextual prescription information, including the reasons for medication use and conditions being treated, achieved the best performance (AUC 0.61–0.63). In contrast, models using only basic drug names or provider information performed poorly (AUC 0.46–0.51). This performance difference was statistically significant ($t = 14.98$, $p < 0.0001$). Our findings suggest that medications data, when analyzed with attention to clinical context, could serve as a low-cost tool for identifying individuals at risk of AD. This approach may help address diagnostic disparities in settings with limited access to advanced testing.